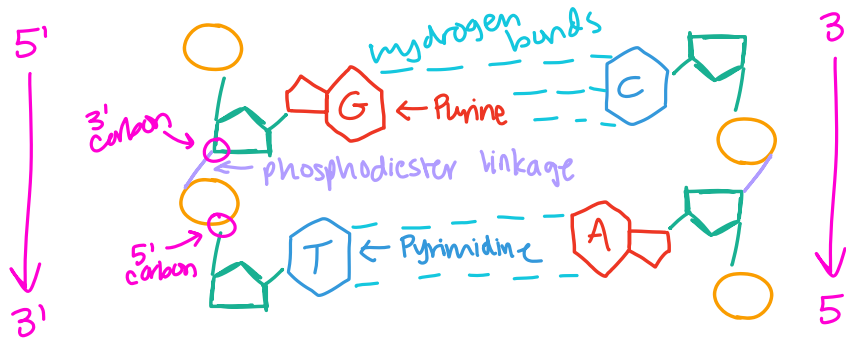


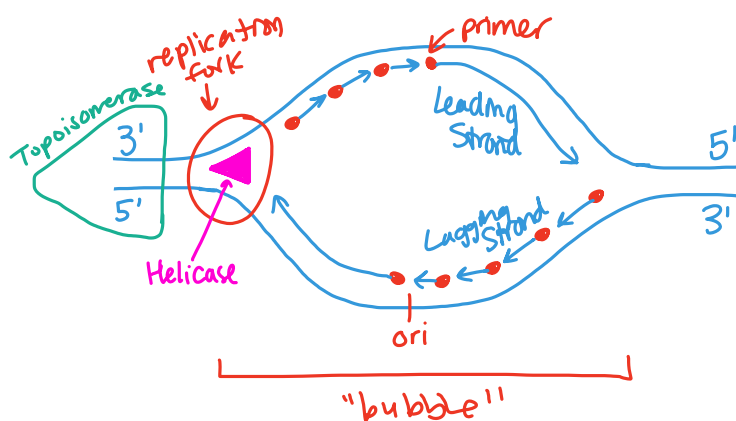
unit 6

- **DNA (Deoxyribonucleic acid)**—the macromolecule that holds genetic information and determines heredity and traits
 - **Nucleotides**—building blocks of DNA, composed of a ringed nitrogenous base, a phosphate group, and a deoxyribose sugar
 - **Purine (Adenine, Guanine)**—double-ringed base
 - **Pyrimidine (Cytosine, Uracil, Thymine)**—single-ringed base
 - **Base-Pairing Rules**—Adenine pairs with Thymine/Uracil, Guanine Pairs with Cytosine. Purine binds to Pyrimidine.
 - **Phosphodiester linkage**—covalent bond between the 3' and 5' end of a nucleotide.
 - **5' (phosphate) end**—connects to the 5' carbon
 - **3' (hydroxyl) end**—connects to the 3' carbon



- **Hydrogen bonds**—weak bonds between nitrogenous bases. Able to be broken apart by helicase. 3 hydrogen bonds between C and G, 2 between A and T.
- **Antiparallel Double-Stranded Sugar-Phosphate Backbone**—polynucleotides running in opposite directions, held together by covalent bonds. One strand runs 5' to 3', and the other 3' to 5'.
- **Double Helix (Watson, Crick, Franklin)**—DNA's shape is of a double helix 
- **Prokaryotic DNA**—circular, singular
- **Eukaryotic DNA**—multiple, linear
- **Plasmids**—double-stranded DNA found in bacteria, prokaryotes, and some eukaryotes that provides genetic advantages and can be manipulated with recombination
- **RNA (Ribonucleic Acid)**—used for transcription and translation, can be the genetic material in some organisms, uses uracil
 - **Uracil**—swapped for Thymine in RNA
 - **Single-Stranded**—RNA typically is single-stranded.
 - **Ribose sugar**—sugar used in RNA but not in DNA
- **DNA Replication**—the process of duplicating DNA for cell division or repair
 - **Parent Strand**—strand that acts as a template during replication
 - **Semiconservative Model**—the DNA unwinds, and each strand is used as a template for another.

- ▶ **Origins of Replication (ori)**—where the DNA strands begin to separate in both directions. There can be multiple ori in eukaryotic DNA replication.
- ▶ **Replication “Bubble”**—the entire area where DNA is actively replicating
- ▶ **Replication Fork**—at the ends of the “bubble,” the Y-shaped region where DNA is separating, forking
- ▶ **Helicase**—enzyme that unwinds the parent strands at the replication fork
- ▶ **Topoisomerase**—the enzyme that uncoils the strands upstream of the replication fork
- ▶ **DNA Ligase**—glues Okazaki fragments together on the lagging strand
- ▶ **Leading Strand**—synthesized continuously, following the replication fork
- ▶ **Lagging Strand**—synthesized discontinuously, away from the replication fork
- ▶ **Okazaki Fragments**—fragmented parts of the lagging strand due to the discontinuous nature
- ▶ **DNA Polymerase**—reads the parent strand in the 3' to 5' direction and synthesizes complementary new nucleotides in the 5' to 3' direction. Also has a proofreading function.
- ▶ **Primase**—adds RNA primers for each strand for DNA polymerase to begin synthesizing a new strand (just one for the leading strand, multiple for each of the Okazaki fragments in the lagging strands)
- ▶ **Proofreading**—DNA Polymerase and other proteins can fix faulty base pairs if there is a problem.

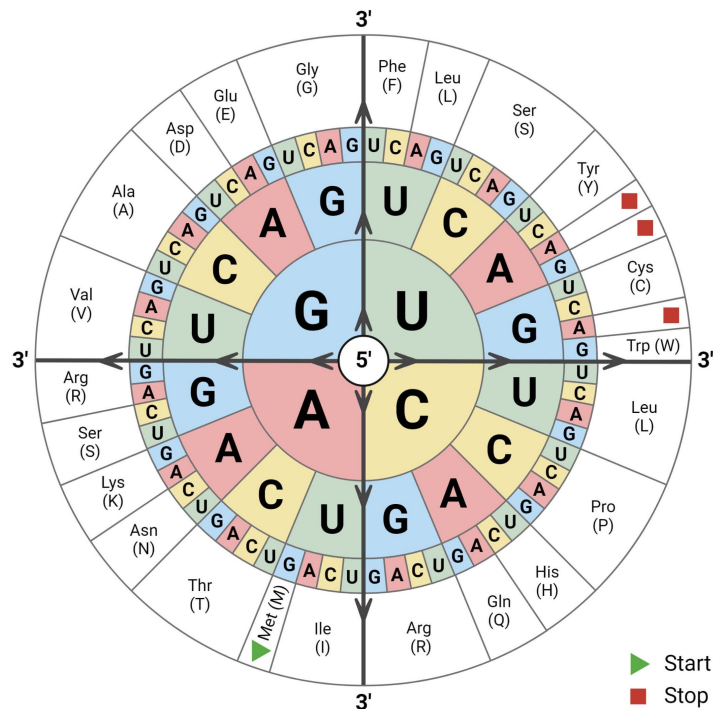


- **RNA**—can hold genetic information, for example—>retroviruses
- **mRNA**—messenger RNA and carries a transcription of DNA from the nucleus to ribosomes for translation, found in all cells
- **rRNA**—ribosomal RNA makes up the structure of the ribosomes due to its folding, found in all cells
- **tRNA**—transfer RNA carries amino acids and binds to codons during translation using anticodons, found in all cells
- **miRNAs**—bind to mRNA to block transcription/translation
- **Central Dogma**—DNA—>RNA—>Protein
- **Transcription**—the process of transcribing mRNA from DNA so that a protein can be made through translation

- **RNA Polymerase**—enzyme that transcribes mRNA from DNA, reads 3' to 5', synthesizes 5' to 3' until it reaches termination sequence
- **Termination Sequence**—sequence of DNA at which the RNA polymerase stops transcribing
- **Promoter**—segment of DNA that RNA polymerase attaches to and starts transcribing from
 - *** “TATA” box in eukaryotes
- **Transcription factors**—mediate binding of RNA polymerase to promoter
- **Transcription Initiation Complex**—transcription factors and promoter and RNA polymerase all working together
- **Template/Noncoding/Antisense Strand**—the strand of DNA that RNA polymerase finds complementary bases for and reads and transcribes from
- **Nontemplate/Coding/Sense Strand**—the strand of DNA that RNA polymerase does not bind to and read and transcribe from
- **Primary Transcript/pre-mRNA**—mRNA before the addition of a poly-A tail or GTP cap in eukaryotes and alternative splicing and removal of introns
- **Poly-A Tail**—added to the 3' end of an mRNA transcript
- **GTP cap**—added to the 5' end of an mRNA transcript
- **Alternative Splicing/RNA Splicing**—sometimes removing certain exons in the final mRNA transcript for alternate proteins from the same DNA while removing all introns
 - *** Prokaryotes do not perform alternative splicing
- **Exons**—coding regions of mRNA
- **Introns**—noncoding regions of mRNA
- **Nuclear pore**—where an mRNA transcript travels through to exit the nucleus
- **Ribosome**—the organelle that manufactures proteins from mRNA transcripts/ instructions, found in the cytoplasm and on the Rough ER in Eukaryotes
- **Genetic Code**—universal code for amino acids that nearly every organism abides by. The genetic code is redundant, but not ambiguous, meaning that every codon will code for a specific amino acid every time, but more than one codon can code for the same amino acid.
- **Amino Acid**—There are 20 amino acids that are synthesized in the ribosomes.
- **Translation**—ribosome reads an mRNA transcript and assembles a protein
- **Initiation**—first step of translation. A tRNA molecule will find the “AUG” codon with its “UAC” anticodon and bind to it, adding the start amino acid, methionine. All following tRNAs will bind according to the position of the start codon, creating a “reading frame.”
- **Reading Frame**—the sets of triplets (codons) that nucleotides are read in. The reading frame can shift due to mutations.
- **tRNA**—carries an anticodon to bind to the mRNA on one side, and an amino acid on the other to add to a growing polypeptide chain.
- **Polypeptide Chain**—the assembled line of amino acids as translated by tRNA
- **Anticodon**—a set of three nucleotides that binds to a complementary set of three mRNA nucleotides (codon)
- **Codon**—any group of three mRNA nucleotides in the mRNA transcript in the reading frame
- **Elongation**—the second step of translation. Amino acids are continuously added after the start codon until a stop codon is reached.

- **Start Codon (AUG)**—initiates translation
- **Stop Codon (UAA, UAG, UGA)**—stops translation
- **Termination**—the last step of translation. Amino acids are no longer added to the polypeptide chain as tRNA detects a stop codon.

Amino Acid Codon Wheel



- **Coupling Transcription and Translation**—prokaryotes can simultaneously translate an mRNA sequence while it is still being transcribed because they lack a nuclear membrane.
*** mRNA is not modified in prokaryotes. It is in eukaryotes.
- **Gene Expression**—the process by which a gene is transcribed, translated, and expressed
- **Functional Gene Product**—the product translated from transcribed mRNA that expresses a gene. Mostly proteins, but can be functional RNA.
- **Gene Regulation**—the process by which living systems can affect how much or to what extent each gene is expressed. This is a dynamic process that changes throughout a living organism's lifetime due to the environment or other pressures.
- **Regulatory Genes**—encode for regulatory proteins that affect transcription (and thus expression)
- **Regulatory proteins**—proteins that bind to promoter sequences to promote or inhibit transcription
- **Small Regulatory RNAs (sRNAs)**—bind to mRNA to inhibit or activate translation (transcription has already happened)
- **Regulatory sequences**—sequences of DNA that regulatory proteins bind to to influence transcription
- **Negative Control**—repressors inhibit transcription when activated
- **Repressor**—a protein that binds to a regulatory sequence to inhibit transcription when activated

- **Positive Control**—activators stimulate transcription when activated
- **Activator**—a protein that binds to a regulatory sequence to stimulate transcription

(The following section is ONLY about bacteria.)

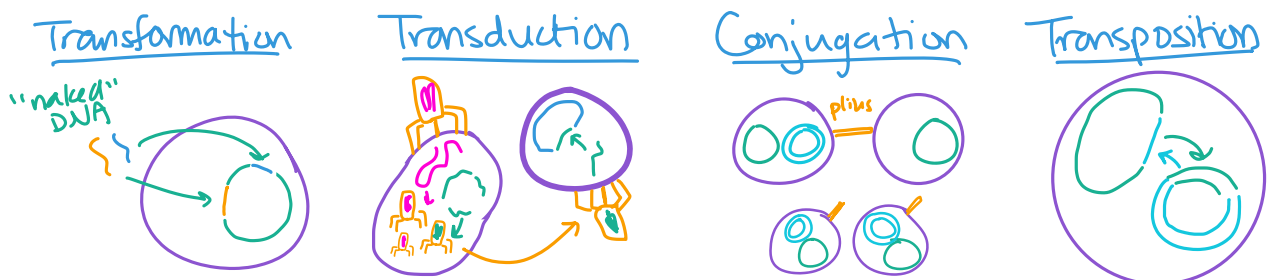
- **Operator**—on-off switch usually within the promoter that proteins can bind to to toggle transcription on or off based on cellular needs. When repressors bind, transcription is inhibited.
- **Promoter**—where RNA polymerase binds to to initiate transcription.
- **Operon**—promoter, operator, and all of the genes they control together
- **Repressible Operons**—gene expression is on by default. A repressor is inactive unless a corepressor is bound to the repressor, activating it. Then, and only then, can the repressor bind to the operator and turn it off. Typically function in anabolic pathways.
 - **trp operon**—tryptophan synthesis is typically on unless trp acts as a corepressor, binding to the repressor that binds to the operator controlling trp synthesis, so that tryptophan is not OVERproduced.
 - **Corepressor**—binds to repressor to activate it
- **Inducible Operons**—operon gene expression is by default inactive, as the repressor is by default active. An inducer can bind to the repressor, inactivating it, so that the repressor can no longer bind to the operator and the gene can be transcribed. Typically function in catabolic pathways.
 - **Inducer**—binds to repressor to inactivate it
 - **lac operon**—allolactose forms as lactose enters a bacterial cell. Allolactose acts as an inducer, binding to the repressor for the lac operon, so that catabolic enzymes can be created and lactose can be broken down.
- **Positive Control (Bacteria)**—regulatory protein/activator, when activated, binds to the operator to stimulate, or increase, gene expression
- **Continuous Gene Expression**—genes are always expressed and always needed

(The following section is ONLY about Eukaryotes.)

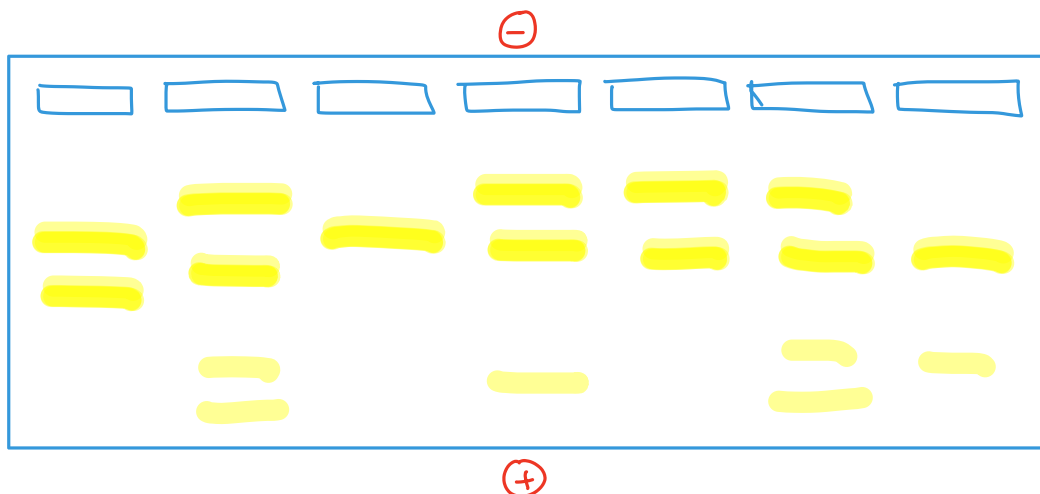
- **Differentiation/Cell Specialization**—leads to cells with particular functions in multicellular organisms, meaning that certain genes are always expressed that are not expressed in other cells, and certain genes are never expressed that are expressed in other cells.
- **Chromatin**—DNA-protein complexes
- **Histones**—proteins chromatin is wound around
- **Nucleosome**—group of a histone and the DNA wound around it
- **Euchromatin**—groups of loosely coiled chromatin (transcriptionally active)
- **Heterochromatin**—tightly coiled chromatin (transcriptionally inactive)
- **Acetylation**—acetyl groups are attached to histone tails, loosening chromatin structure
- **Methylation**—methyl groups are attached to DNA bases (typically cytosine) to coil the DNA more tightly and thus reduce transcription
- **Epigenetics**—the environment changing gene expression (phenotype) without altering genes (genotype), such as coiling of chromatin and transcriptional activity
- **Enhancers**—regulatory sequences that activators bind to to increase transcription

- **Transcription Factors**—bind to promoters to ensure that RNA polymerase can bind; basically activate transcription
- **General Transcription Factor**—required for gene expression of all protein-coding genes
- **Specific Transcription Factor**—activators/repressors that bind to a specific regulatory sequence for a specific gene to influence transcription
 - *** Eukaryotes, because they do not have operons and thus do not have a system where multiple genes share a promoter and operator, can use the same transcription factors for groups of related genes to couple transcription of certain proteins/products.
 - ▶ **Activator (Eukaryotes)**—binds to an enhancer to increase transcription
 - ▶ **Enhancer**—where an activator binds to increase transcription
 - ▶ **Repressor (Eukaryotes)**—binds to a silencer sequence to decrease transcription
 - ▶ **Silencer**—where a repressor binds to decrease transcription
- **Post-Transcriptional Regulation**—gene expression is fine-tuned even after transcription.
 - ▶ **Alternative Splicing**—different combinations of exons produce different proteins
 - ▶ **RNA Interference (RNAi)**—miRNAs (microRNAs) bind to mRNA to inhibit transcription and degrade the mRNA
- **Mutations**—ultimate source of all genetic variation, as mutations create changes to DNA sequences.
 - ▶ **Somatic Mutation**—occurs in a body cell and is not passed to offspring
 - ▶ **Germ Line Mutation**—occurs in gametes/germ line cells and is passed to offspring
 - ▶ **Germ Line Cell**—somatic cell that will become a gamete
 - ▶ **Base-Pair (bp)**—one set of complementary nucleotides in DNA
 - ▶ **Point Mutations**—changes to just one base-pair (bp) of a gene, which can produce a very different protein
 - **Base-Pair Substitution**—replacing one bp with another
 - **Silent Mutation**—the transcribed mRNA still encodes for the same amino acid, and thus the same protein.
 - **Missense Mutation**—the transcribed mRNA encodes for a different amino acid, changing the protein.
 - **Nonsense Mutation**—the final protein is smaller than intended because of a premature stop codon. Almost always nonfunctional.
 - **Frameshift Mutations**—shift the “reading frame” and thus affect every amino acid after the mutation
 - **Insertion**—a bp is inserted
 - **Deletion**—a bp is deleted
 - ▶ **Spontaneous Mutation**—occurs accidentally and randomly with no outside or environmental influence
 - ▶ **Mutagen**—physical or chemical agents that cause mutations (chemical substances, UV radiation...)

- ▶ **Environmental Context**—mutations provide alternate alleles and thus increase genetic diversity (beneficial), can cause a silent mutation where there is no effect and thus no natural selection (neutral), and can cause loss of function or genes needed for survival (harmful).
- **Sexual Reproduction**—increases genetic variation through fertilization, crossing over, law of segregation, and independent assortment
 - ▶ **Fertilization**—a genetically diverse egg and sperm create a completely unique and novel combination of DNA.
 - ▶ **Crossing Over**—Homologues exchange genetic information to create new chromosomes with different combinations of genes
 - ▶ **Law of Segregation**—an individual can pass on either of two alleles, for every gene
 - ▶ **Independent Assortment**—each pair of Homologues lines up independent of each other during metaphase I of Meiosis I and thus are separated independently of which parent they came from
- **Asexual Reproduction**—prokaryotes simply duplicate. Provides no genetic diversity.
- **Horizontal Gene Transfer**—acquiring new genetic information from an organism that an individual is relatively unrelated to in prokaryotes
 - ▶ **Transformation**—uptake of “naked” DNA from a nearly lysed cell and possible incorporation into the cell’s genome
 - “naked” DNA—unprotected
 - ▶ **Transduction**—foreign bacterial DNA is introduced to a new cell by a bacteriophage (virus). When a bacteriophage infects a host cell, it typically fragments the host genome while it inserts its own genetic information for creating more viruses. Sometimes, these fragments can get loaded into retroviruses, which then go infect other cells. However, instead of infecting the viral DNA into the cell, it gives the cell the DNA of another host, which could then be incorporated into the current host’s genome to increase genetic diversity.
 - ▶ **Conjugation**—two cells in direct contact exchange (through a pilus) a plasmid, allowing a new cell to grow a pilus and have the plasmid
 - **Plasmid**—double-stranded DNA that provides genetic diversity and viral resistance
 - **Pilus**—bridge-like structure connecting two prokaryotes
 - ▶ **Transposition**—segments of DNA from one DNA molecule relocate to another. This commonly happens between the cell’s genome and its plasmid.



- **Virus**—organisms that infect a host cell to create more viruses. Their genetic information can be single-stranded or double-stranded, DNA or RNA. They have greater faster genetic variation than most organisms because of recombination, enzyme use, and reproduction speed.
 - **Recombination**—two viruses infect the same cell, meaning their genetic information gets packaged together
 - **RNA Polymerase Use**—RNA viruses use RNA-dependent RNA Polymerase, which leads to higher rates of mutation (no proofreading function) and thus greater genetic variation.
 - **Reproduction**—viruses reproduce faster, meaning there are more viruses and greater risk of mutation, especially with an enzyme with no proofreading function.
- **Retrovirus**—a virus that inserts its genetic information (RNA) into a host cell to be incorporated as DNA in the host genome, breaking the central dogma. (RNA→DNA instead of DNA→RNA)
- **Reverse Transcriptase**—the enzyme retroviruses use to turn their injected RNA into DNA so that it can be incorporated into the host genome.
- **Biotechnology**—manipulating and analyzing DNA and RNA
- **Polymerase Chain Reaction (PCR)**—amplifying a DNA sample through artificially-controlled cell division and high temperatures
 - **Taq polymerase**—a thermostable enzyme used for DNA replication in PCR. Does not denature at the high temperatures PCR requires.
 - **Denaturation**—heat up to denature DNA, separating DNA strands
 - **Annealing**—cool so that primers can bind to the single-stranded DNA
 - **Extension**—warm up slightly so Taq polymerase can extend the primers and fully replicate the DNA strands
- **Gel Electrophoresis**—uses a restriction enzyme to cut DNA or RNA at certain sequences and then runs an electrical current to separate molecules (DNA, RNA, proteins) by size and charge. DNA is negatively charged and is thus attracted to the positive cathode.
 - **Restriction Enzyme**—found in bacteria. Cuts DNA/RNA at a specific nucleotide pattern (ex: GGCACT)
 - **Crispr-Cas9**—uses RNA sequence to guide to a complementary sequence of DNA/RNA, and then cuts using the enzyme.
 - **Banding Patterns**—the result of gel electrophoresis. How far a specific band (same size + charge molecules) traveled is determined by how large and how charged they were. How thick or clear the band is indicates how many molecules are in that band.



*** Gel Electrophoresis can be used to determine one's DNA fingerprint, parent, or genotype

- **Bacterial Transformation**—recombinant DNA plasmids can introduce new DNA into bacterial genome. First, recombinant plasmids are made by cutting in specific spots in the DNA and then on the plasmid, and then inserting the DNA into the gap in the plasmid. Ligase glues the fragments together. Then, transformation will move a gene from a plasmid to the host genome, where it will be transcribed and translated to produce a new protein!
- **DNA Sequencing**—During DNA Replication, “chain-terminating,” fluorescently labeled nucleotides (ddNTPs) can terminate the DNA strand and indicate which nucleotide base is at which position. DNA Polymerase, while replicating, may use a ddNTP, terminating the fragment. Gel electrophoresis will then sort these fragments by size, indicating that the fragments that travel further and terminate earlier and thus are shorter will start the sequence.

